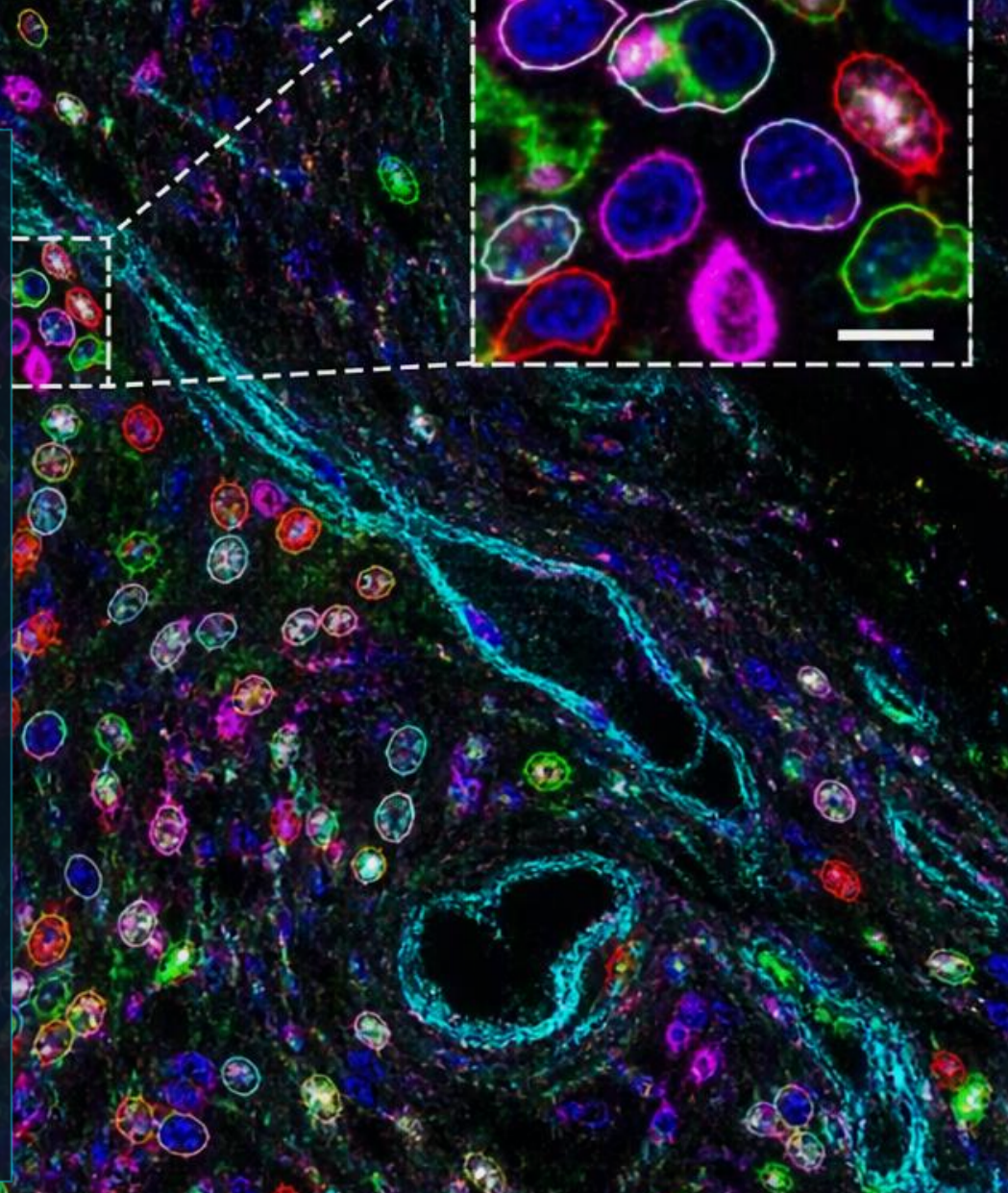


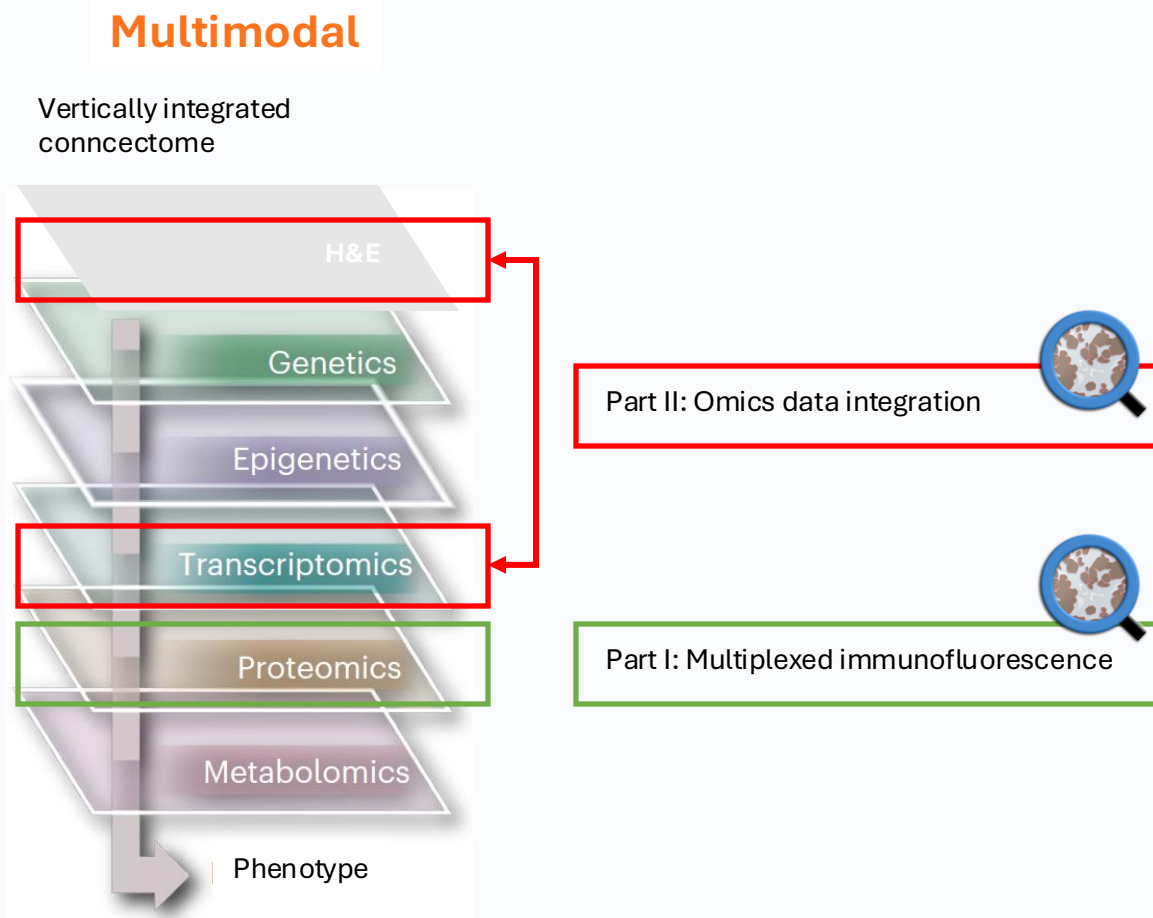
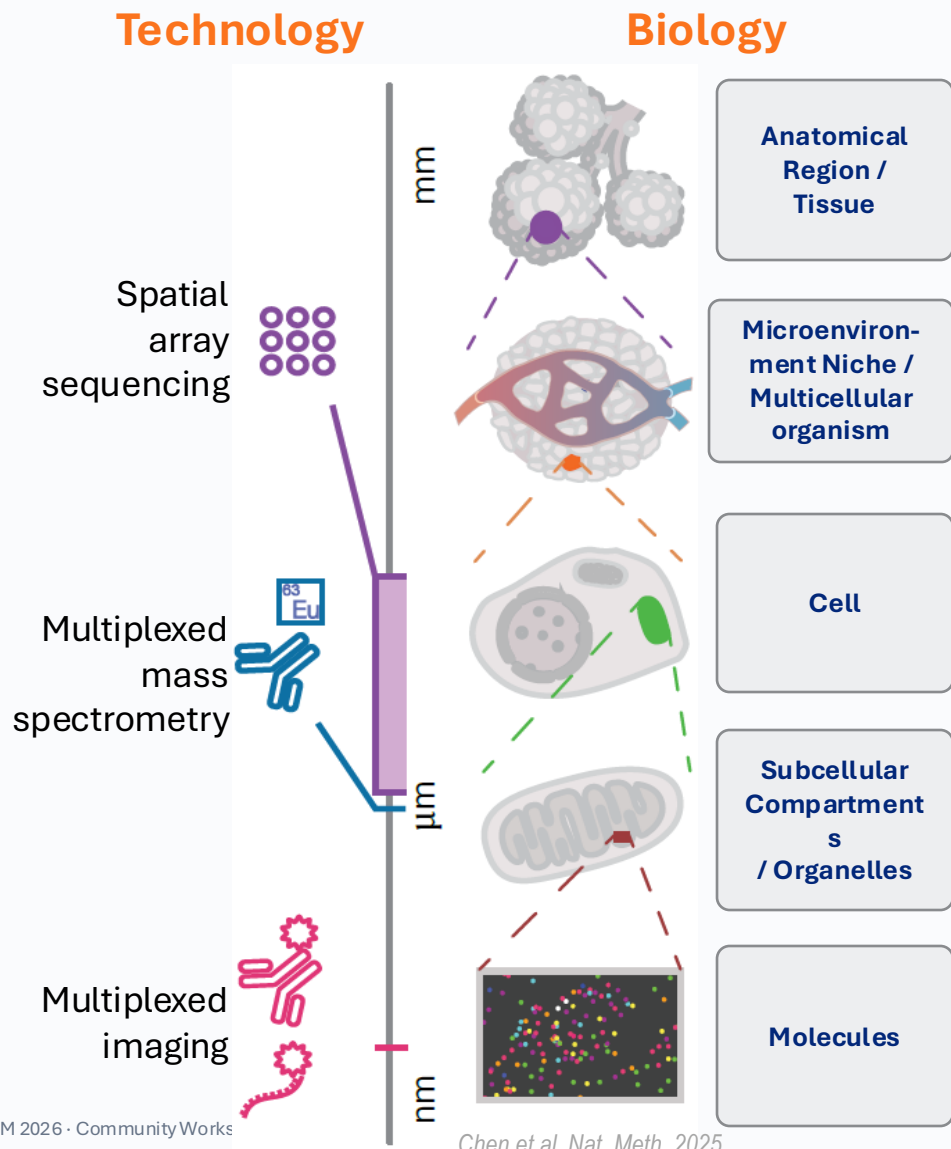
QuPath: A great tool for spatial omics image and data analysis

Part I · Multiplexed Immunofluorescence Image Analysis

Community Workshop · SPAOM 2026
Vic · 6–9 October 2026
CNIO · Confocal Microscopy Unit



Two Complementary Routes into Spatial Multiomics



Multiplex and High-Plex Imaging

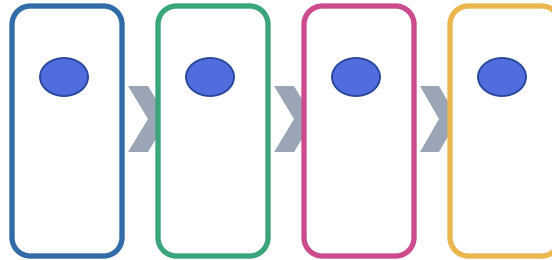
Many markers measured in the same tissue section while preserving architecture



MULTIPLEX

Several markers measured in the same tissue section

Benefit: direct co-localization



C1

C2

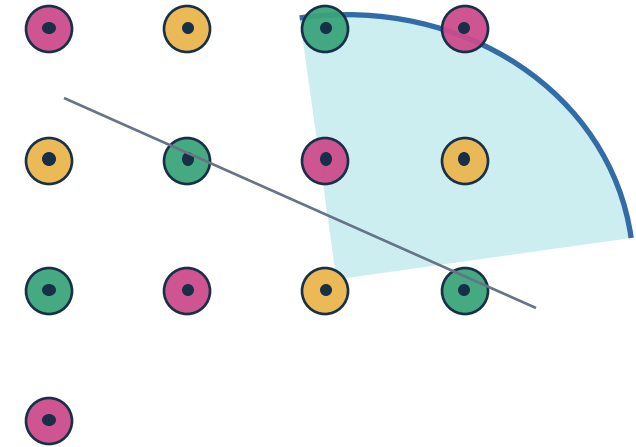
C3

C4

HIGH-PLEX

Dozens of markers, frequently acquired over repeated cycles

Risk: cycle-to-cycle variation



SPATIAL CONTEXT

Every measurement remains linked to its location in tissue architecture

Biological meaning: who is near whom?

Plex describes how many targets are measured; spatial context gives those targets biological meaning.

Multiplex measures several targets; high-plex increases target number; spatial analysis preserves relationships to tumor, stroma and neighboring cells.

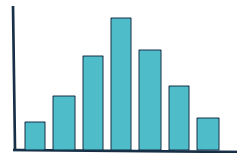
How are Multiplex Images Acquired?

Different technologies create different artefacts, that must be controlled before biological interpretation **but the downstream logic is shared**



SIMULTANEOUS FLUORESCENCE

Multiple fluorophores acquired together



Strengths

- Fast
- Direct composite

Watch for

- Spectral overlap
- Autofluorescence
- Panel design



CYCLIC FLUORESCENCE

Stain → image → remove or inactive signal → repeat



Strengths

- Higher plex
- Standard optics

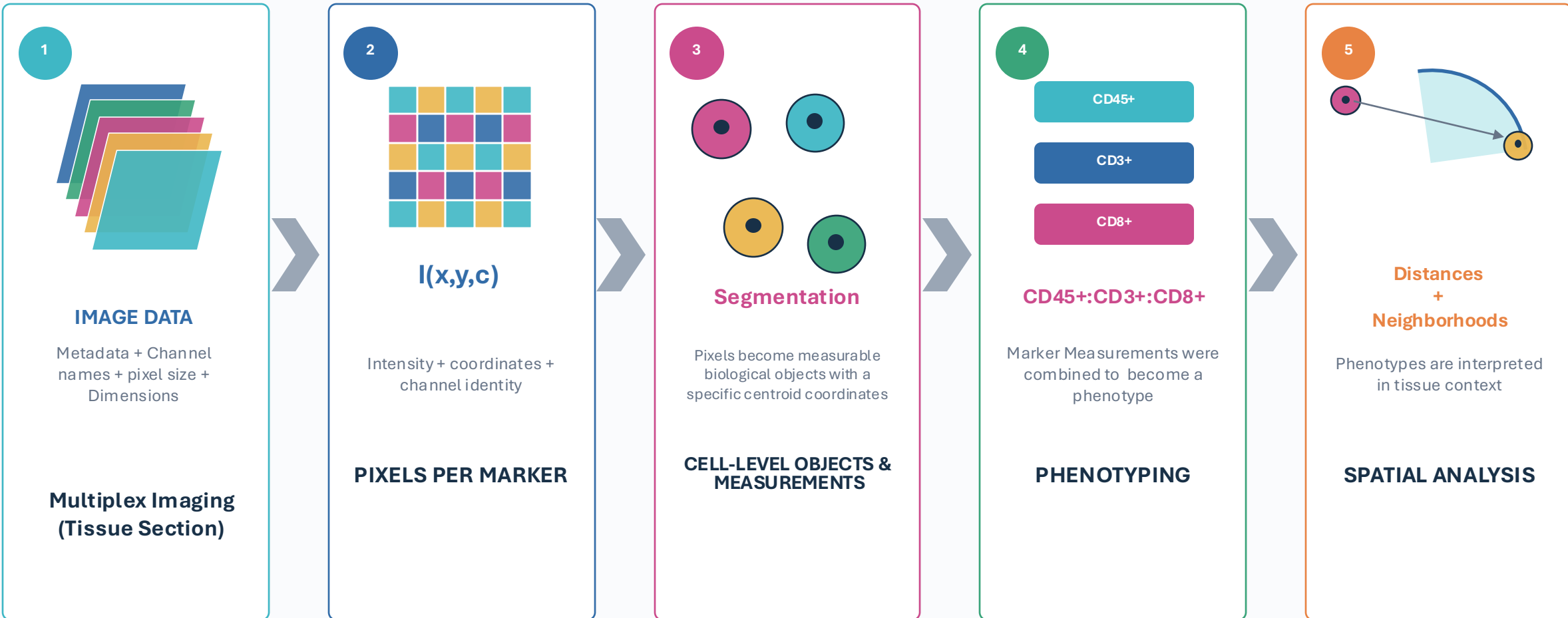
Watch for

- Registration errors
- Cycle QC (cycle consistency)
- Signal consistency
- Tissue loss and deformation

Instrument choice changes the artefacts, but the downstream logic remains similar

From Multiplex Image to Spatial Biology

A single **workflow** connects **pixels**, **cells**, **phenotypes** and **tissue context**



The analysis moves from “Marker Signal” to “Cell identity” to “Tissue Organization”

Why QuPath for Spatial Omics Images?

An open-source environment connecting the image to cell-level data and reproducible analysis



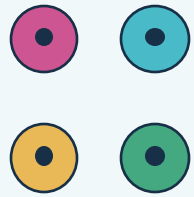
VIEW & NAVIGATION

Navigate large images (whole-slide) and inspect channels (multichannel images)



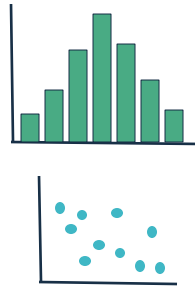
ANNOTATE

Define regions, tissue compartments and training examples.



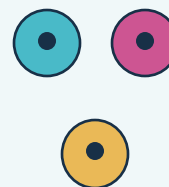
DETECT

Create nuclei, cells, spots or other objects with classical or deep-learning methods



MEASURE

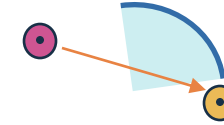
Attach intensity, morphology, textures and XY coordinates at cell level.



Phenotypes

CLASSIFY

Convert single/multiple measurements into Single/Composite classifiers to create biological labels.



ANALYZE

Relate objects to borders (annotations) and neighbors (cell detections).



EXPORT

Save tables, objects, classifiers and scripts for downstream analysis.

QuPath is the bridge from multichannel pixels to spatially located cell objects.

Understanding QuPath Objects

An **object** is something in the **image** that QuPath can **identify**, **classify** and **measure**

Two Main Object Types



ANNOTATIONS

Role: biological context or a parent analysis region

- Usually created by the **user** by drawing a region.
- **Flexible** and **editable**.
- Can contain **annotations** and **detections**
- Examples: Full image, tumor, stroma...



DETECTIONS

Role: Child objects provide cell-level data

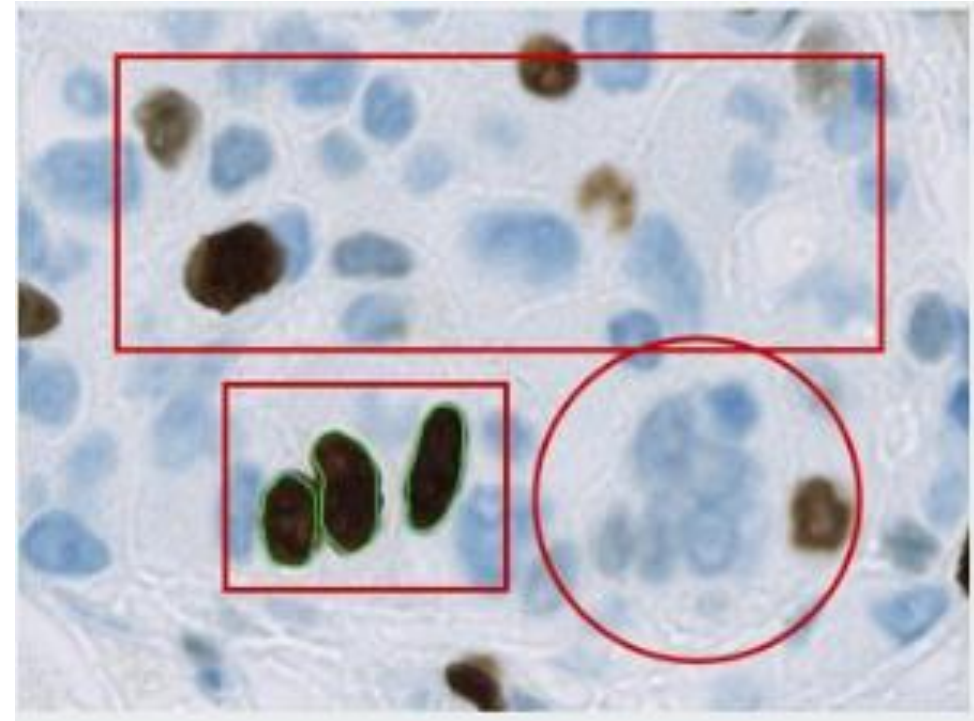
- Usually created **automatically** by an **algorithm**
- Examples: cells, nuclei and subcellular spots

Annotations are flexible

- Usually **tens** rather than thousands
- Can define **ROIs** or **training regions**
- Often define where **detections** are generated

Detections are numerous

- Often **hundreds** to **millions**
- Carry **measurements** and **coordinates**
- Can be deleted, but not editable



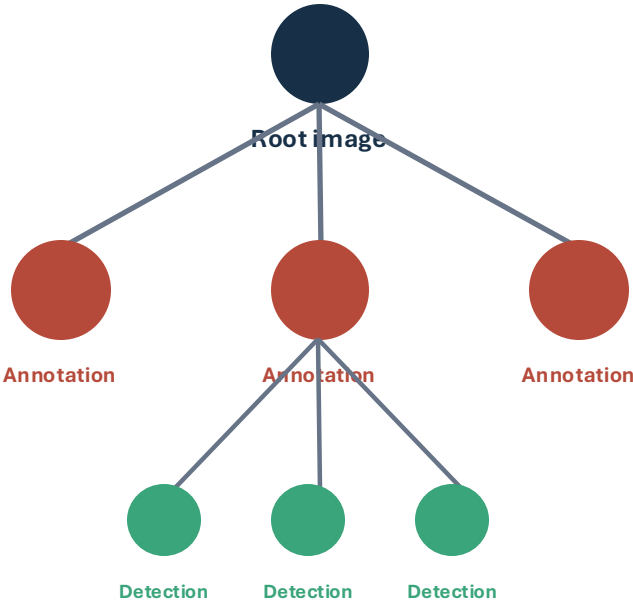
A rectangular annotation containing many cell detections

Practical rule: draw annotations to define context; use detections to quantify individual biological structures.

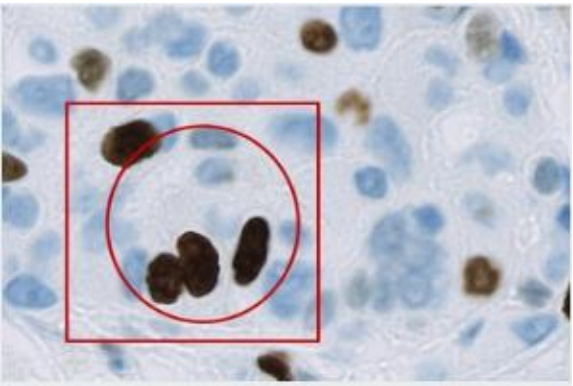
Object Hierarchies: Parent–Child Organization

QuPath organizes **objects** as a **tree**: every **object** has **one parent** and may have **multiple children**

Tree-like Structure

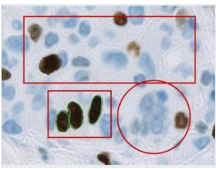


One parent · multiple child objects

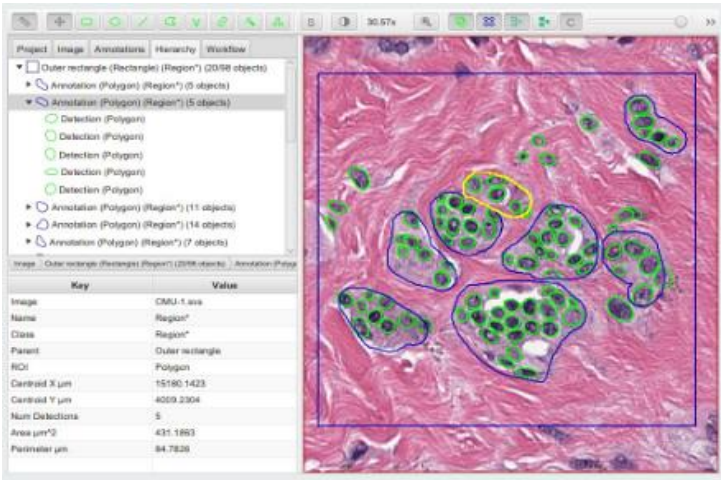


Nested annotations

A rectangular annotation can contain a circular annotation, that contains detections. The inner object becomes a child of the outer region.



Annotation containing detections



Visualizing the Hierarchy

The Hierarchy panel reveals parent–child relationships and helps confirm that detections belong to the intended annotation.

Key consequences

- Selected parents define analysis scope
- Measurements can be summarized by parent
- Deleting a parent may affect child objects

Always confirm the selected parent object before running detection, creating masks or deleting annotations.

Before Analysis: Understand the Image

Six checks prevent technical properties from being mistaken for biology

1



DAPI · CD3 · CK

Channels & Markers

Do names match the panel?
Which channel contains DAPI + protein markers?

2

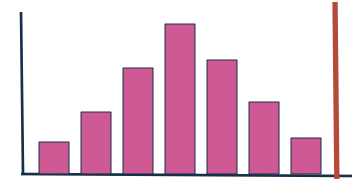


0.5 μm / px

Dimensions, Scale & Calibration

Pixel size converts pixels into μm and controls object size.

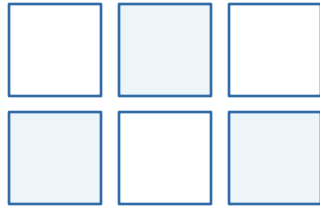
3



Dynamic Range

Check bit depth, background, weak signal and saturated pixels.

4



Geometry

Confirm tiles, cycles, z-planes, orientation and registration.

5



Image Format

OME-TIFF or company formats preserves image data and metadata, but verify both.

6



Quality

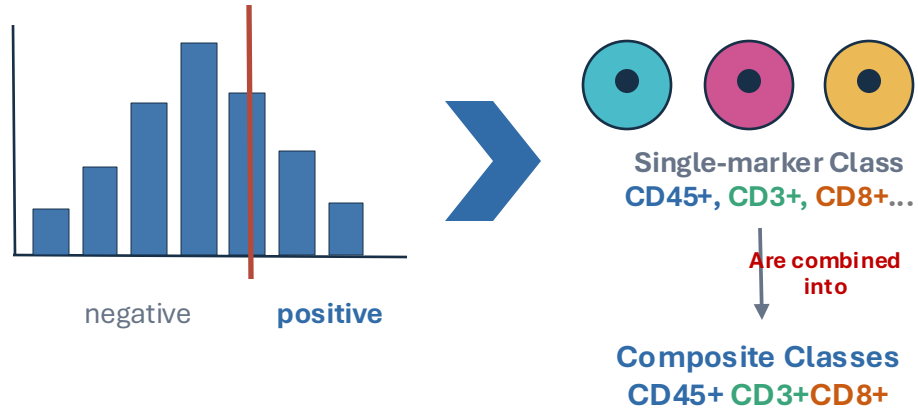
Look for blur, autofluorescence, bleed-through and tissue loss/deformation.

Never start the analysis before checking these 6 points, phenotyping and downstream analysis can be wrong.

Phenotyping: Two Complementary Strategies

Phenotyping converts cell **measurements** into biological **categories**. Both approaches classify the same segmented cell objects

THRESHOLD-BASED CLASSIFICATION



Best when marker distributions are separable and a transparent rule is preferred

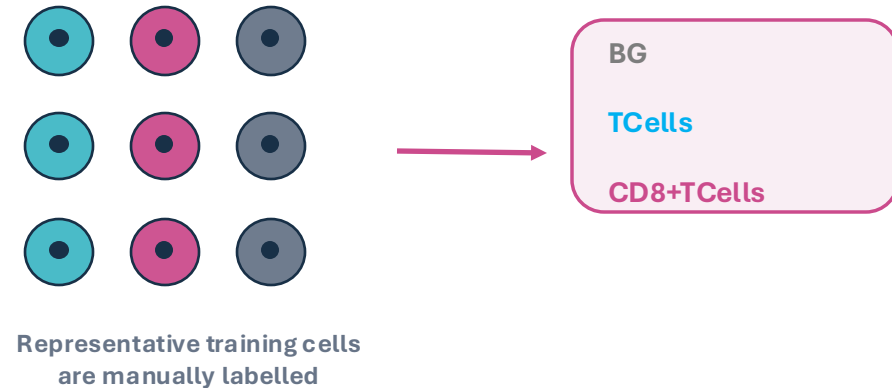
Strengths

- Fast
- Interpretable
- Reproducible

Limitation

- One measurement at a time
- **Thresholds** may not represent **heterogeneous** or **overlapping** populations

SUPERVISED OBJECT ML CLASSIFICATION



Best when multiple measurements help distinguish heterogeneous populations

Strengths

- Handles heterogeneity
- Uses multiple measurements simultaneously
- Life update & iterative refinement

Limitation

- Depends on training quality
- Biased

Threshold-based results are useful guides for selecting training examples, but they are not ground truth.

How a QuPath Composite Classifier Builds Cell Phenotypes

The **composite classifier** executes **several single-measurement classifiers** and **combines the classes** assigned to each cell.

1 · RUN SINGLE CLASSIFIERS

Each classifier evaluates **one measurement** independently.

CD45 classifier

CD45: Cell mean



CD45+

immune criterion

CD3 classifier

CD3: Cell mean



CD3+

T-cell criterion

CD8 classifier

CD8: Cell mean



CD8- or
CD8+

subset criterion

2 · COMBINE ASSIGNED CLASSES

QuPath does not directly search for a predefined “T cell”.

Instead, for every cell:

- A Run each saved single classifier
- B Retain every class assigned to the cell
- C join the class names into one compound class

Possible outputs from the composite classifier

CD45+

CD45+:CD3+

CD45+:CD3+:CD8
-

CD45+:CD3+:CD8
+

3 · INTERPRET COMPLETE COMBINATIONS

Only specific complete compound classes correspond to the target phenotypes used in this workshop.



T cell

CD45+ : CD3+ : CD8-

→ TCells



CD8+ T
cell

CD45+ : CD3+ : CD8+

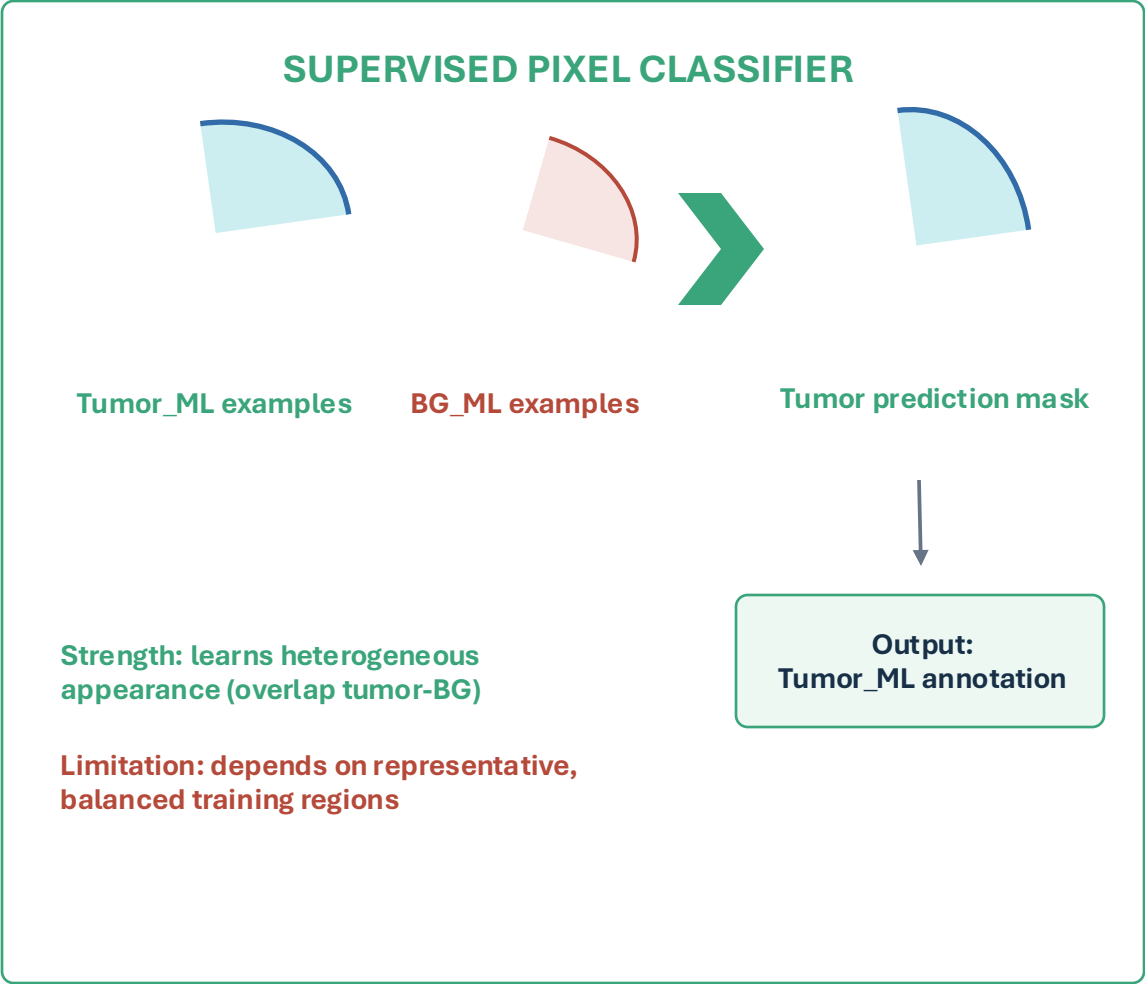
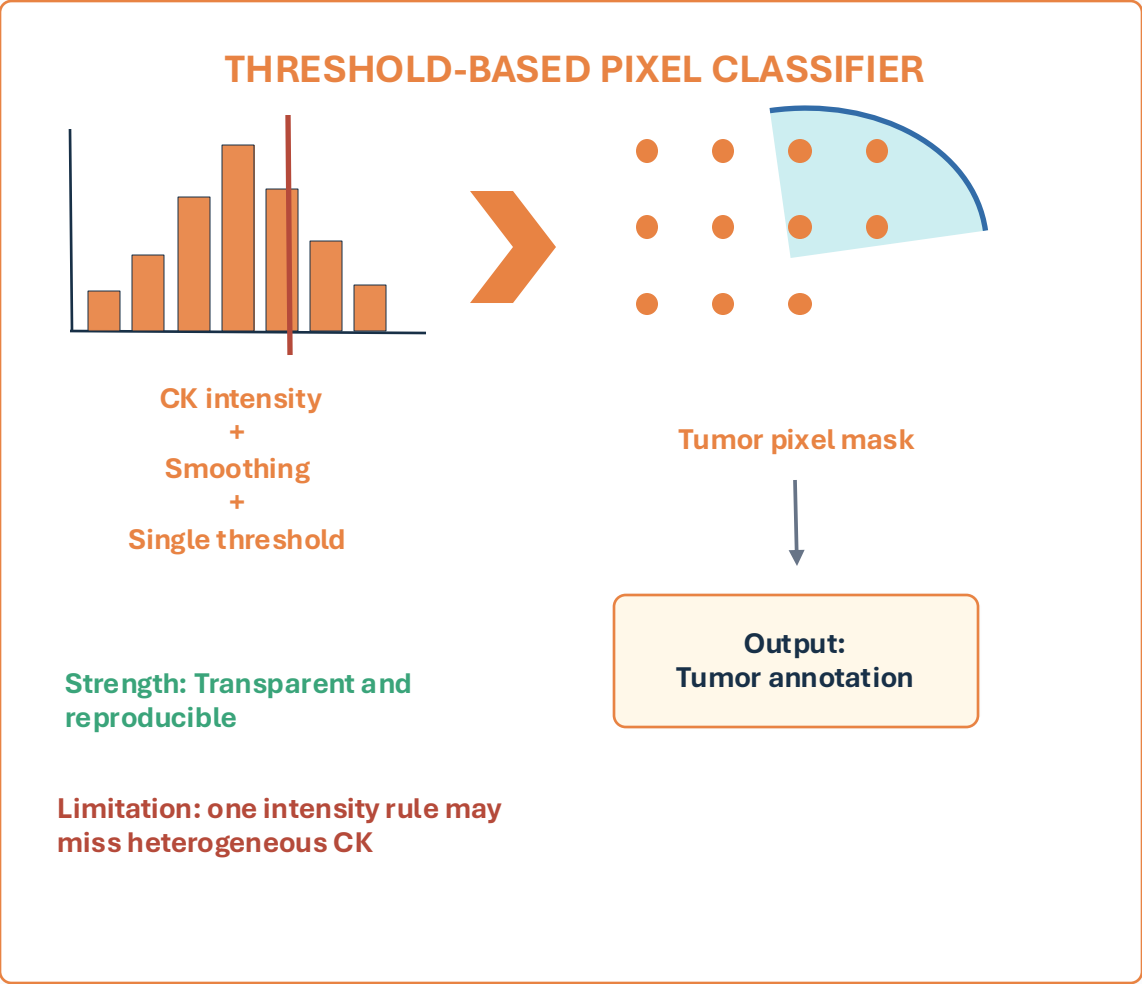
→ CD8+TCells

Intermediate classes are informative: they show which criteria were met before the final biological label.

Composite classifier philosophy: combine the evidence assigned by each single classifier first; interpret selected complete combinations as biological phenotypes second.

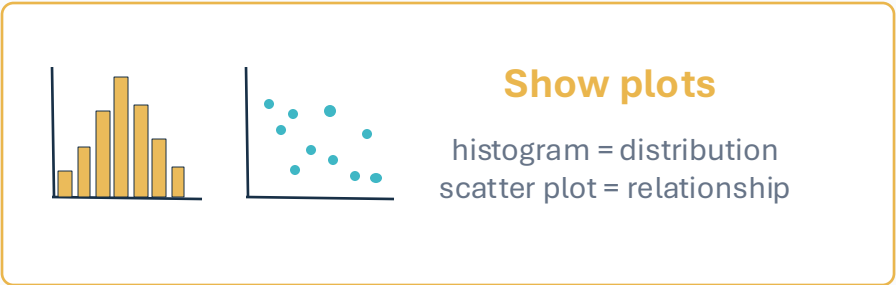
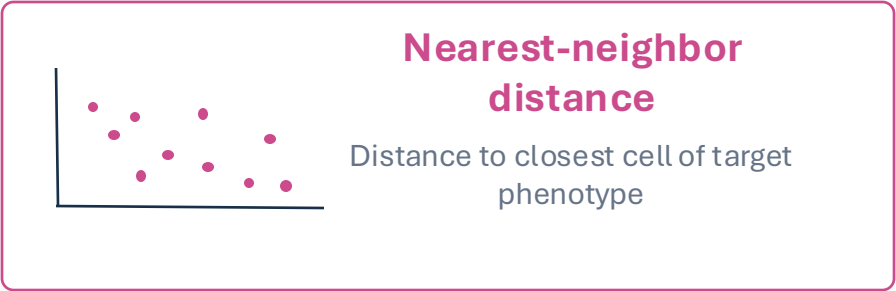
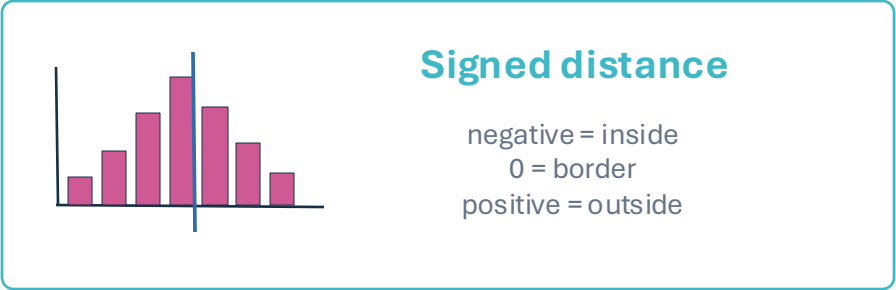
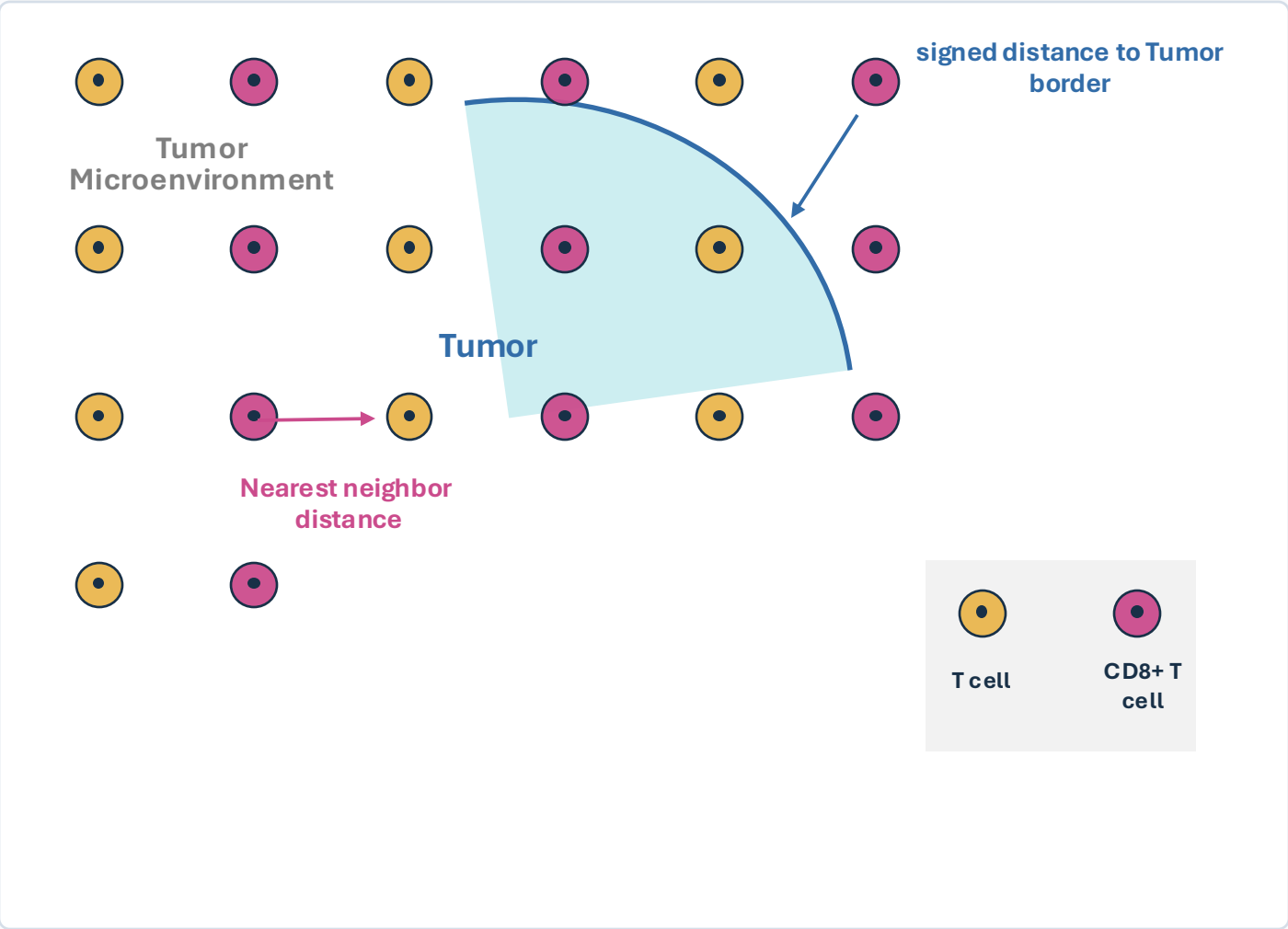
Tumor Compartment: Define the Spatial Reference

The tumor mask answers “where is the tumor?” and becomes the reference for spatial measurements



Spatial analysis adds the “where” to the “what”

Spatial measurements relate classified cells to tissue compartments and to one another

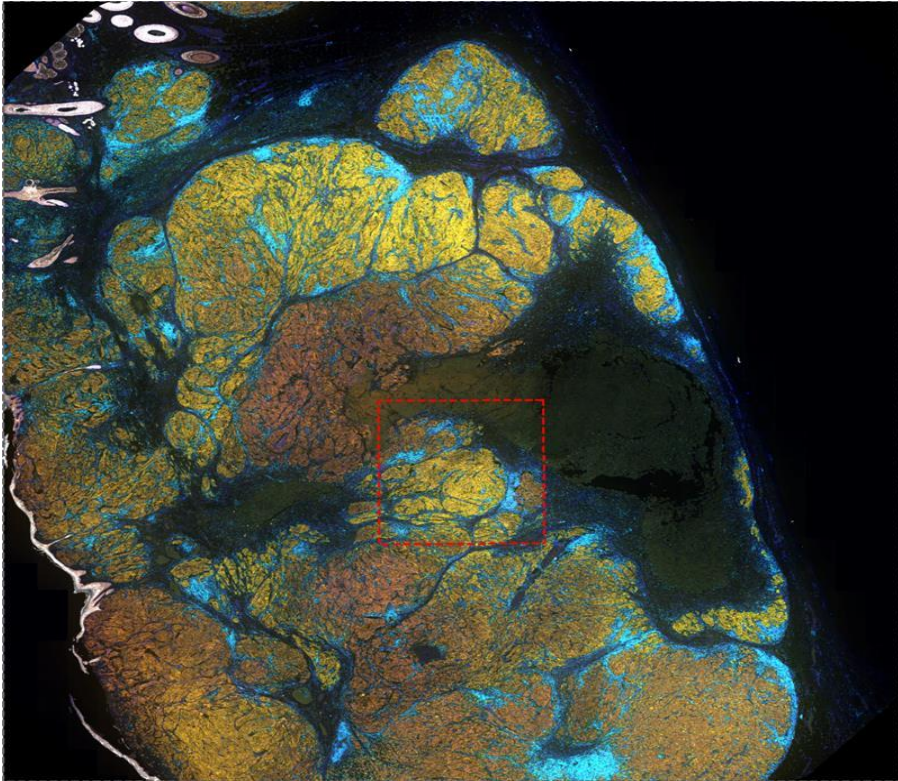


Proximity is descriptive: it does not by itself prove biological interaction.

From the Whole Melanoma Section to the Workshop Crop

Spatial biology starts with tissue-wide architecture, but the hands-on analysis will focus on one representative region.

1 · WHOLE-TISSUE CONTEXT

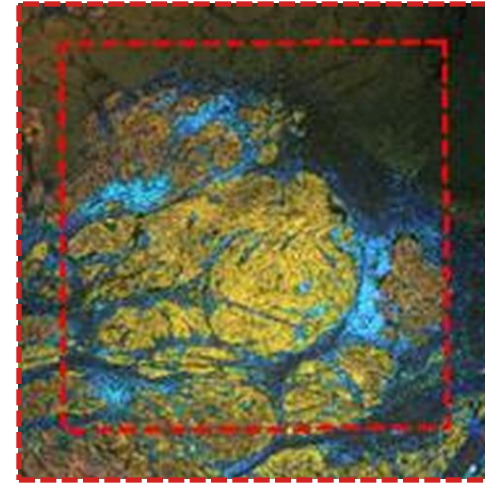


Original multiplex melanoma image



select ROI

2 · PRACTICAL ANALYSIS REGION



Why work on a crop?

- Whole-section segmentation and measurement are computationally demanding.
- A smaller region keeps the workshop reproducible and responsive.
- The same QuPath workflow can later be applied to larger regions or the full image.

What remains spatially meaningful?

Channel intensities · cell boundaries · cell phenotypes · X/Y coordinates · local tissue organization

Interpret results within the selected ROI: the crop is a teaching region, not a summary of the entire tissue section.

Workshop strategy: preserve the biological logic of whole-tissue spatial analysis while reducing the computational scale.

What we will do in the hands-on

The practical exercise reproduces the complete analytical reasoning, not only software clicks

01



Open project

Confirm image type, metadata, paths, channels and scale.

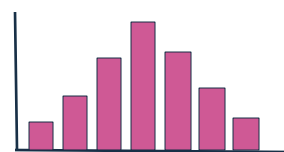
02



Segment cells

Run Cellpose or import pre-generated GeoJSON detections.

03



Measure markers

XY centroid and cell compartments intensity measurements.

04



Phenotype

Compare threshold rules with supervised object ML.

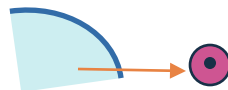
05



Tumor Annotation

Compare CK thresholding with supervised pixel ML.

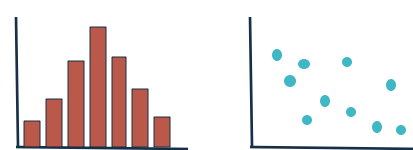
06



Analyze space

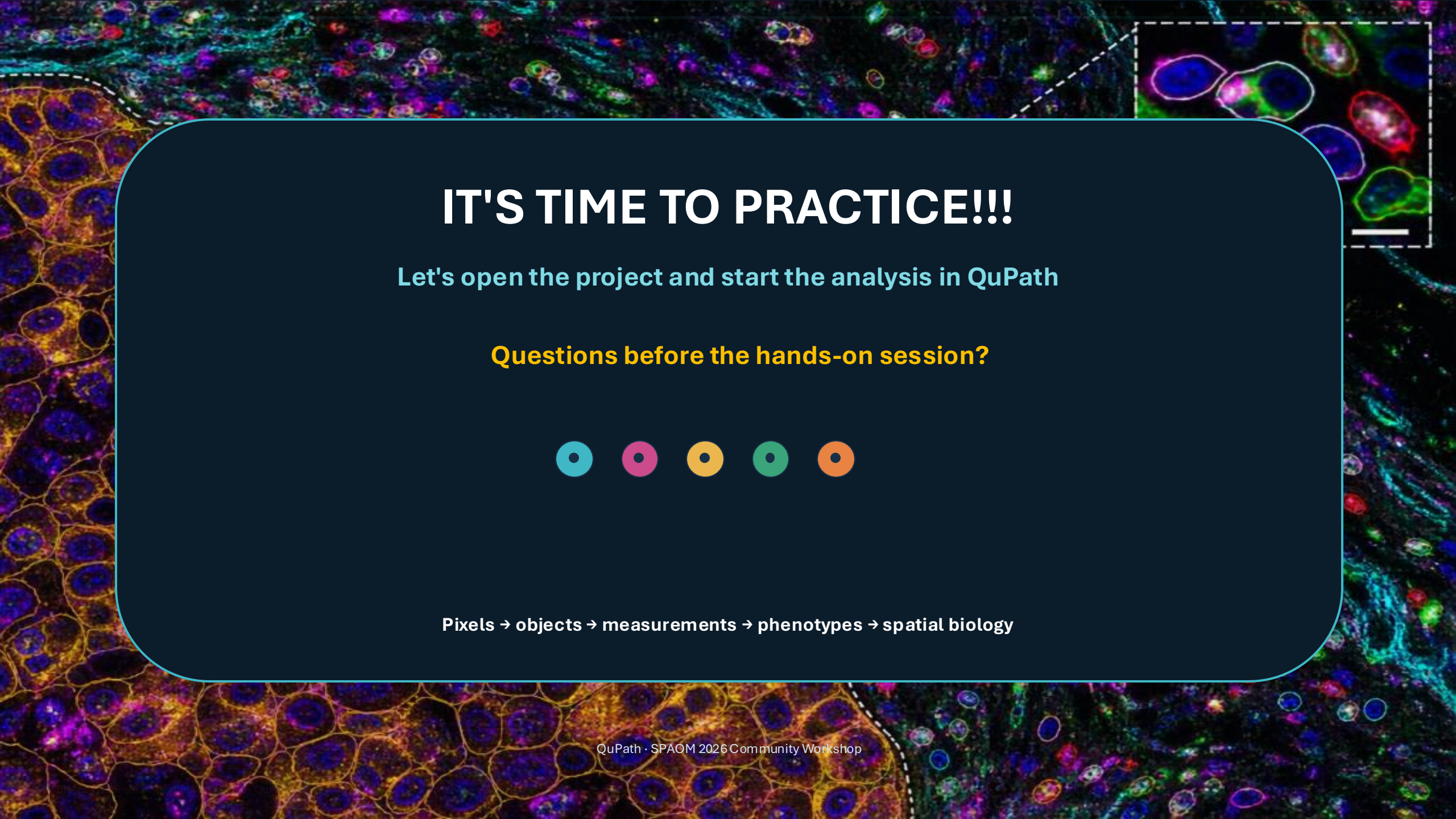
Measure tumor-border and nearest-neighbor distances.

07



Visualize & Export

Use tables, histograms and scatter plots for QC and interpretation.



IT'S TIME TO PRACTICE!!!

Let's open the project and start the analysis in QuPath

Questions before the hands-on session?



Pixels → objects → measurements → phenotypes → spatial biology